Chapter 7
Number theory

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you should be able to do

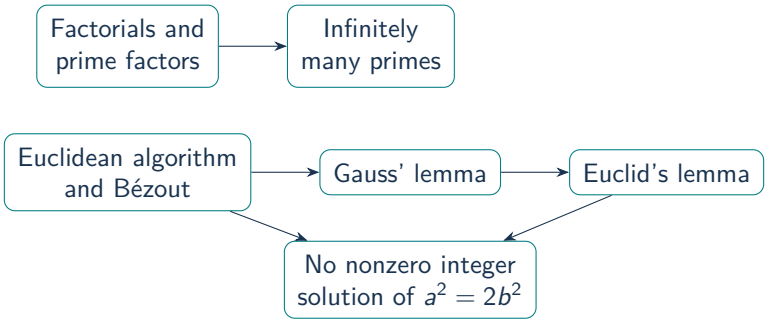
- Explain why a prime divisor of $N! + 1$ must exceed N.
- Use a Bézout identity to justify cancellation in divisibility.
- Derive Euclid's lemma from the defining property of a prime.
- Identify the decreasing positive integer in an infinite-descent proof.
- Connect an integer equation to irrationality.

Capstone perspective

The new results combine familiar tools: witnesses, case splits, induction, gcd, and contradiction. Each proof also needs a specific mathematical construction.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 7, Sections 7.1–7.3.

A map of the ingredients



For the descent proof, the needed parity fact can also be obtained directly from Chapter 6.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 7; prerequisites in Sections 6.2, 6.4, and 6.7.

7.1: What does it mean to have infinitely many primes?

Prime

A natural number p is prime if $p \geq 2$ and each natural divisor of p is either 1 or p.

We will prove the concrete statement

$$\forall N \in \mathbb{N}, \quad \exists p \in \mathbb{N}, \quad p > N \text{ and } p \text{ is prime.}$$

Why is that enough? Any finite list of natural numbers has an upper bound N. The theorem supplies a prime outside that list. The construction will produce a *prime factor* of a new number, rather than require that the new number itself be prime.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1, theorem on infinitely many primes; definition from 4.1.8.

Build a number with a prime factor

Fix an arbitrary $N \in \mathbb{N}$ and put $M = N!$.

1. $M \geq 1$, so $M + 1 \geq 2$.
2. Every natural number at least 2 has a prime factor. Choose a prime p with $p \mid M+1$.
3. Write $M + 1 = pk$ with $k \in \mathbb{N}$.
4. Since $M + 1 > 0$, $k \neq 0$. Write $k = \ell + 1$ with $\ell \in \mathbb{N}$.

We now know

$$p \geq 2, \qquad M + 1 = p(\ell + 1).$$

The next step will prove that p cannot also divide M.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; uses factorial positivity and the prime-factor theorem.

The new prime factor is larger than the bound

From $M + 1 = p(\ell + 1) = p\ell + p$ and $p \geq 2$, we get

$$p\ell < M < p(\ell + 1).$$

Thus M lies strictly between consecutive multiples of p, so $p \nmid M$.

1. Suppose $p \leq N$.
2. Since $1 \leq p \leq N$, the factorial-divisibility theorem gives $p \mid N! = M$.
3. This contradicts $p \nmid M$.

Therefore $p > N$. Since N was arbitrary, there are infinitely many primes.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; uses Examples 4.5.8 and 6.2.5.

The construction can start with a composite number

N	$N! + 1$	Prime factor p	Comparison
4	$25 = 5^2$	5	$5 > 4$
5	$121 = 11^2$	11	$11 > 5$
6	$721 = 7 \cdot 103$	7	$7 > 6$

What the theorem needs

The integer $N! + 1$ only needs to be at least 2 so that it has a prime divisor. Every such prime divisor exceeds N.

The proof gives a new prime beyond a chosen bound; it does not claim that factorial-plus-one is always prime.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; original numerical illustrations.

7.1 Practice: explain the construction's logic

1. Run the proof for $N = 0$. What are $N! + 1$ and its prime factor?
2. Suppose $p_1, \ldots, p_r$ is a nonempty finite list containing every prime. Set

$$K = p_1 p_2 \cdots p_r + 1.$$

   Explain why a prime divisor of K cannot belong to the list.

You may use: if an integer divides two integers, it divides their difference.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; classroom extensions of its construction.

7.1 Solution ideas

1. $0! = 1$, so $0! + 1 = 2$. The prime 2 is larger than 0; no special exception is needed.
2. $K \geq 2$, so choose a prime $q \mid K$. If $q = p_i$ for some listed prime, then q divides $p_1 \cdots p_r$ as well. Hence
$$q \mid (K - p_1 \cdots p_r) = 1,$$
which is impossible for $q \geq 2$.

The factorial version excludes every prime up to N; the product version excludes every prime in the given list.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; solutions to the preceding extensions.

7.2: When may a factor be cancelled in divisibility?

Gauss' lemma

Let $a, b, d \in \mathbb{Z}$. If

$$d \mid ab \qquad \text{and} \qquad \gcd(a, d) = 1,$$

then $d \mid b$.

The condition $\gcd(a, d) = 1$ says that a and d are coprime.

Why the hypothesis matters

$6 \mid 2 \cdot 3$, but $6 \nmid 3$. Here $\gcd(2, 6) = 2$, so the coprimality assumption fails.

Bézout's identity will express 1 as a combination of a and d.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Gauss' lemma; counterexample is an extension.

Gauss' lemma: construct the divisibility witness

Choose $x, y \in \mathbb{Z}$ with $xa + yd = 1$, using Bézout and $\gcd(a, d) = 1$. Choose $z \in \mathbb{Z}$ with $ab = dz$, using $d \mid ab$. Then

$$\begin{aligned} b &= b(xa + yd) \\ &= x(ab) + byd \\ &= x(dz) + byd \\ &= d(xz + by). \end{aligned}$$

Thus $xz + by \in \mathbb{Z}$ witnesses $d \mid b$.

Lean connection

`obtain` extracts the Bézout coefficients and divisibility witness; `use` supplies the new witness; `rw` and `ring` verify the calculation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, proof of Gauss' lemma.

A concrete cancellation problem

Let $n \in \mathbb{Z}$ and suppose $35 \mid 12n$. Prove $35 \mid n$.

1. The identity $3 \cdot 12 - 35 = 1$ supplies the needed coprimality information.
2. Write $12n = 35k$ with $k \in \mathbb{Z}$.
3. Multiply the identity by n:

$$\begin{aligned} n &= 3(12n) - 35n \\ &= 3(35k) - 35n \\ &= 35(3k - n). \end{aligned}$$

The proof not only shows divisibility; it gives a quotient $3k - n$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Gauss' lemma; original worked instance.

Euclid's lemma: primes divide one factor

Theorem

For $a, b, p \in \mathbb{N}$, if p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Put $d = \gcd(a, p)$. We know $d \mid a$ and $d \mid p$. Since p is prime, $d = 1$ or $d = p$.

Case	Conclusion
$d = 1$	Gauss' lemma applied to $p \mid ab$ gives $p \mid b$.
$d = p$	Since $d \mid a$, we have $p \mid a$.

Primality reduces the gcd to exactly the two cases needed by the proof.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Euclid's lemma.

A prime divisor of a power divides the base

Let $a, p, k \in \mathbb{N}$, with p prime and $k \geq 1$. Then

$$p \mid a^k \quad \Longrightarrow \quad p \mid a.$$

Prove this by induction on k, starting at 1.

- **Base:** $a^1 = a$.
- **Step:** if $p \mid a^{t+1} = a \cdot a^t$, Euclid's lemma gives $p \mid a$ or $p \mid a^t$. The first case finishes; the second uses the inductive hypothesis.

A consequence we will use

If a^2 is even, then a is even: apply the result with the prime $p = 2$ and $k = 2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, power corollary.

Lean keeps the integer and natural-number statements separate

The textbook defines its gcd on $\mathbb{Z}$, while its prime predicate is on $\mathbb{N}$.

1. Cast a and p to integers to apply gcd and Gauss' lemmas.
2. Use $\gcd(a, p) \geq 0$ to represent that integer gcd by a natural number d.
3. Transfer the divisibility facts to $\mathbb{N}$, where the prime definition applies.

Textbook notation/tactic	Purpose
`(a : Int)`	View a natural number as an integer.
`lift`	Replace a nonnegative integer by a natural number.
`norm_cast`	Transport compatible statements across casts.

The mathematical case split remains $d = 1$ or $d = p$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Lean discussion following Euclid's lemma.

7.2 Practice: choose the correct divisibility tool

1. Let $n \in \mathbb{Z}$ and $9 \mid 4n$. Use $7 \cdot 4 - 3 \cdot 9 = 1$ to produce a witness for $9 \mid n$.
2. Let $u, v \in \mathbb{N}$. If $7 \mid u^2 v$ and $7 \nmid u$, prove $7 \mid v$.
3. Find a composite d and natural a, b such that $d \mid ab$, but $d \nmid a$ and $d \nmid b$.

Name every primality or coprimality hypothesis used in your proof.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2; original classroom exercises.

7.2 Solution ideas

1. Write $4n = 9k$. Then
$$n = 7(4n) - 27n = 9(7k - 3n).$$
So $7k - 3n$ is a witness.
2. Euclid's lemma gives $7 \mid u^2$ or $7 \mid v$. The first case implies $7 \mid u$ by the power corollary, contradicting the hypothesis. Hence $7 \mid v$.
3. Take $d = 6$, $a = 2$, and $b = 3$. This shows why Euclid's lemma requires a prime divisor.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2; solutions to the preceding exercises.

7.3: The arithmetic statement behind irrationality

Theorem

There are no $a, b \in \mathbb{N}$ with $b \neq 0$ such that $a^2 = 2b^2$.

The assumption $b \neq 0$ is essential: $(a, b) = (0, 0)$ satisfies the equation. We will prove by strong induction on b the property

$$P(b): \quad \forall a \in \mathbb{N}, \quad b \neq 0 \Rightarrow a^2 \neq 2b^2.$$

Why quantify over all numerators?

The next solution will have a different numerator. The induction hypothesis must cover every a at each smaller denominator.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, natural-number theorem and descent proof.

From one hypothetical solution, build a smaller one

Suppose $a, b \in \mathbb{N}$, $b \neq 0$, and $a^2 = 2b^2$.

1. $2 \mid a^2$, so $2 \mid a$. Write $a = 2k$ for some $k \in \mathbb{N}$.
2. Substitute and cancel the nonzero factor 2:

$$2b^2 = a^2 = (2k)^2 = 4k^2 \quad \Longrightarrow \quad b^2 = 2k^2.$$

3. Thus (b, k) satisfies the same form of equation as (a, b).

The remaining obligations

To use the induction hypothesis, we must prove $k \neq 0$ and $k < b$. A new solution alone does not yet give a contradiction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, natural-number descent; parity fact also follows from 7.2.

Check the denominator really decreases

We have $b \neq 0$ and $b^2 = 2k^2$.

1. If $k = 0$, then $b^2 = 0$, hence $b = 0$, a contradiction. Thus $k > 0$.
2. Since $k^2 > 0$,
$$k^2 < 2k^2 = b^2.$$
Because $k, b \geq 0$, this implies $k < b$.
3. The strong-induction hypothesis at denominator k, with numerator b, says $b^2 \neq 2k^2$.

This contradicts the new equation. Therefore the assumed original solution cannot exist.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, well-foundedness lemma and natural-number proof.

The precise role of the induction hypothesis

At denominator b, the hypothesis covers *all* pairs (r, s) with $s < b$:

$$\forall r, s \in \mathbb{N}, \quad s < b \Rightarrow s \neq 0 \Rightarrow r^2 \neq 2s^2.$$

Once the preceding slides supply $k < b$, $k \neq 0$, and $b^2 = 2k^2$:

```
-- Local demonstration fragment in the induction step:
-- IH covers every numerator r at every denominator s < b.
-- hkb : k < b; hk : Not (k = 0)
-- hnew : b ^ 2 = 2 * k ^ 2
have forbidden := IH b k hkb hk
exact forbidden hnew
```

The textbook expresses this as a recursive theorem call, with termination measured by the second argument.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, theorem irrat_aux and its termination explanation.

A second proof: use gcd and Bézout

Now let $a, b \in \mathbb{Z}$ with $b \neq 0$, and suppose $a^2 = 2b^2$. Put $d = \gcd(a, b)$.

1. Since $d \mid a$ and $d \mid b$, choose $k, \ell \in \mathbb{Z}$ with

$$a = dk, \qquad b = d\ell.$$

2. Bézout gives $x, y \in \mathbb{Z}$ with $ax + by = d$.
3. $d \neq 0$: otherwise $b = d\ell = 0$, a contradiction.

We will construct an integer z with $z^2 = 2$, which is impossible.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, integer theorem and Bézout proof; notation normalized.

The key identity creates an integer square root of 2

Set $z = 2\ell x + ky \in \mathbb{Z}$. Then

$$\begin{aligned} d^2 z^2 &= (2bx + ay)^2 \\ &= 2(ax + by)^2 + (y^2 - 2x^2)(a^2 - 2b^2) \\ &= 2d^2 + (y^2 - 2x^2) \cdot 0 \\ &= 2d^2. \end{aligned}$$

The middle equality is a polynomial identity, verified by expansion. Since $d \neq 0$, cancellation of d^2 gives $z^2 = 2$.

Where each input was used

$a = dk$, $b = d\ell$: first line. $ax + by = d$: third line. $a^2 = 2b^2$: third line. $d \neq 0$: cancellation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, integer proof using Brahmagupta's identity.

Why the integer square equation is impossible

If $z \in \mathbb{Z}$, then $|z|$ is a nonnegative integer. Exactly one of the following ranges applies:

Range	Bound on z^2
$\lvert z\rvert \leq 1$	$z^2 = \lvert z\rvert^2 \leq 1 < 2$
$\lvert z\rvert \geq 2$	$z^2 = \lvert z\rvert^2 \geq 4 > 2$

Thus $z^2 \neq 2$ for every integer z, contradicting the constructed z.

Conclusion of the second proof

There are no integers a, b with $b \neq 0$ and $a^2 = 2b^2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, completion of the integer proof; prerequisite Example 2.3.5.

Extension: translating the equation into irrationality

A real number is *rational* when it can be written a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$. A real number that is not rational is *irrational*. Suppose a rational $q = a/b$ satisfied $q^2 = 2$.

Multiplication by b^2 would give

$$a^2 = 2b^2,$$

contradicting the integer theorem.

Consequence

Every real solution of $q^2 = 2$ is irrational. In particular, the positive solution $\sqrt{2}$ is irrational.

No assumption that a/b is in lowest terms is needed by these proofs.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, arithmetic theorem; rational-number interpretation added for CSC302.

7.3 Practice: audit a descent argument

1. In the natural-number proof, explain why the induction hypothesis must allow a numerator different from the original a.
2. A proposed proof produces another solution (a', b') but only shows $b' \leq b$. Explain why this does not finish a descent argument.
3. Adapt the descent proof to show that no natural a, b with $b \neq 0$ satisfy $a^2 = 3b^2$. You may use Euclid's lemma for the prime 3.

For item 3, list the new pair and prove both required inequalities about its denominator.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3; original proof-audit and generalization exercises.

7.3 Solution ideas

1. The new pair is (b, k), so the numerator changes from a to b. The induction property is quantified over every numerator.
2. A process may repeat the same denominator forever. Strong induction requires a *strictly* smaller nonnegative integer.
3. From $3 \mid a^2$, obtain $a = 3k$. Then

$$a^2 = 3b^2 \Rightarrow 9k^2 = 3b^2 \Rightarrow b^2 = 3k^2.$$

Here $k \neq 0$ because $b \neq 0$, and $k^2 < 3k^2 = b^2$ gives $k < b$. The pair (b, k) contradicts the induction hypothesis at k.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3; solutions to the preceding exercises.

Review: the construction and the reason it works

Result	Construction	Decisive fact
Infinitely many primes	Prime divisor of $N! + 1$	It cannot divide $N!$.
Gauss' lemma	Multiply a Bézout identity by b	An explicit multiple of d results.
Euclid's lemma	Examine $\gcd(a, p)$	A prime has only two natural divisors.
No $a^2 = 2b^2$, $b \neq 0$	Smaller solution, or an integer square root of 2	Strict descent, or the gap between 1 and 4.

In every proof, identify the witness, preserve its domain, and check the assumption that permits cancellation or induction.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 7, Sections 7.1–7.3.

Sources and further work

Heather Macbeth, *The Mechanics of Proof*, Chapter 7.

- 7.1: factorial construction for infinitely many primes.
- 7.2: Gauss' lemma, Euclid's lemma, and the power corollary.
- 7.3: the natural-number descent and integer Bézout proofs.

Read Chapter 7 and compare the mathematical and Lean proofs.

The prose, diagrams, and classroom exercises in this deck are original adaptations. The rational-number interpretation is an explicit extension. Lean fragments use the textbook's educational environment.